DATA CLEANING AND EDA ON HOUSING DATA

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INTRODUCTION

- **Objective**: Perform EDA on raw_house_data.csv.
- **Purpose**: Identify key trends and insights to understand the dataset better.
- **Methods Used**: Data Cleaning, Visualization, and Statistical Analysis.

DATASET OVERVIEW

- This dataset is structured similarly to real estate prediction datasets, where the goal is to analyze property attributes to understand pricing trends.
- **Number of Rows**: 5000 (Each row represents a single house listing)
- **Number of Columns**: 16 (Each column represents a key attribute of a house)
- **Target Variable**: **sold_price** (Represents the final selling price of the house)

Variable	Description	raw_house_data.csv
MLS	Unique identifier for each property listing	✅
sold_price	Final selling price of the property	✅
zipcode	ZIP code where the property is located	✅
longitude	Longitude coordinate of the property	✅
latitude	Latitude coordinate of the property	✅
lot_acres	Size of the lot in acres	✅
taxes	Property taxes in USD	✅
year_built	Year the property was constructed	✅
bedrooms	Number of bedrooms in the property	✅
bathrooms	Number of bathrooms in the property	✅
sqrt_ft	Total square footage of the property	✅
garage	Number of garage spaces	✅
kitchen_features	Features available in the kitchen (e.g., Dishwasher, Refrigerator)	✅
fireplaces	Number of fireplaces	✅
floor_covering	Types of floor coverings used in the property	✅
HOA	Homeowners Association fees (if applicable)	✅

DATA CLEANING AND EDA

After importing necessary python libraries like pandas, numpy seaborn and matplotlib and loading csv file

```
data=pd.read_csv('/content/drive/MyDrive/DS_Assignment1/raw_house_data - raw_house_data.csv')
```

Data cleaning and preprocessing is a crucial step in Data Science for model prediction. It involves identifying and correcting errors in the dataset by handling missing values, removing duplicates, and fixing inconsistencies. It ensures the data is accurate, consistent, and ready for analysis or machine learning.

Steps Involved include:

1. **Handling missing values**
2. **Exploratory Data Analysis**
3. **Feature Engineering**

4.1 HANDLING MISSING VALUES

Under **Data Cleaning & Preprocessing**, the **Handling Missing Values** step ensures data completeness by addressing gaps in the dataset.

For our given **raw_house_data.csv**

1. checked for any missing values as first step. data.isnull().values.any()

2. I filled all numerical columns with **mean** and all categorical columns with **mode.**

3. Converted and cleaned few numeric columns and drop rows with missing values.

selected_features = ['sold_price', 'sqrt_ft', 'bedrooms', 'bathrooms', 'garage', 'lot_acres', 'taxes','HOA']

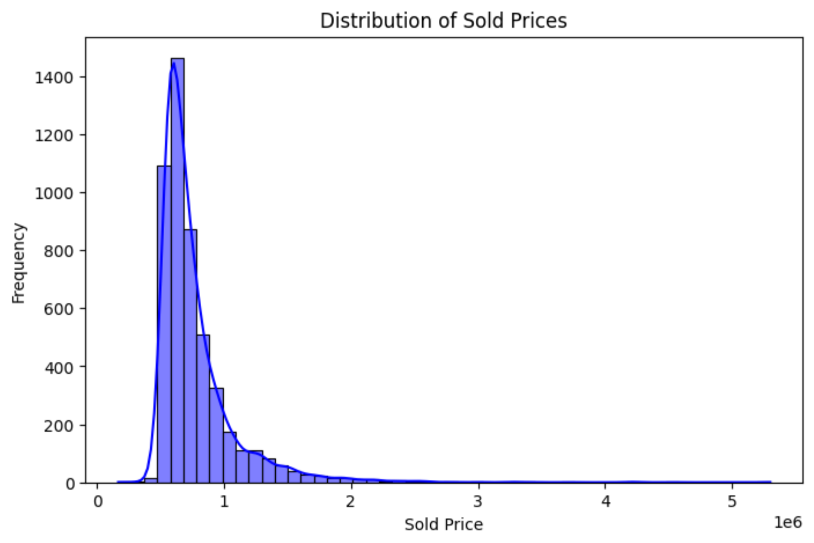
4. After handling missing values we finally got the shape of dataset

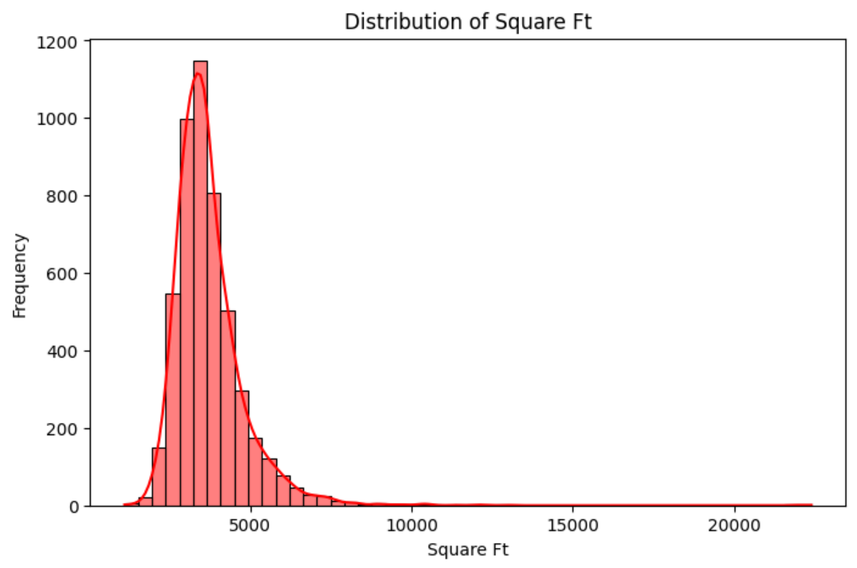
```
[ ] data.shape

(4982, 16)
```

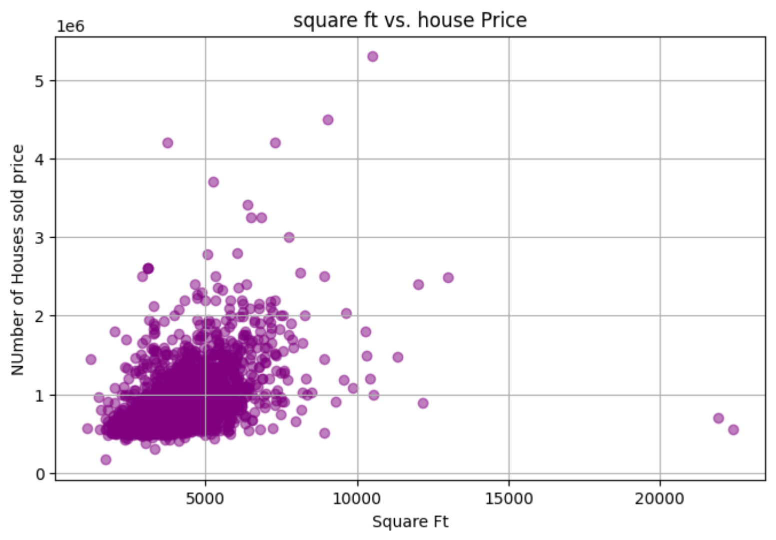
4.2 EXPLORATORY DATA ANALYSIS

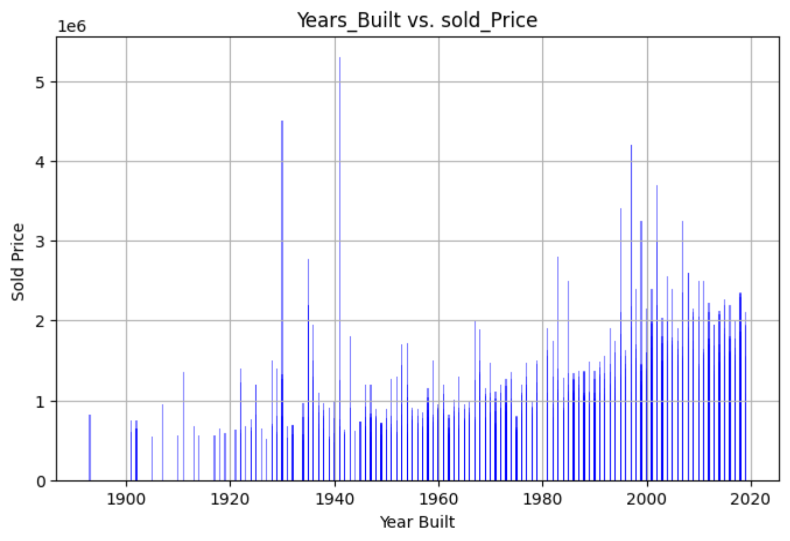
- After handling missing values and describing the dataset **data.describe**() and displaying basic info **data.info**()
- Now, will explore every features, so we can and select only those data-columns which has significance to classify the target variable which is **sold _price** in our case and plot visualizations

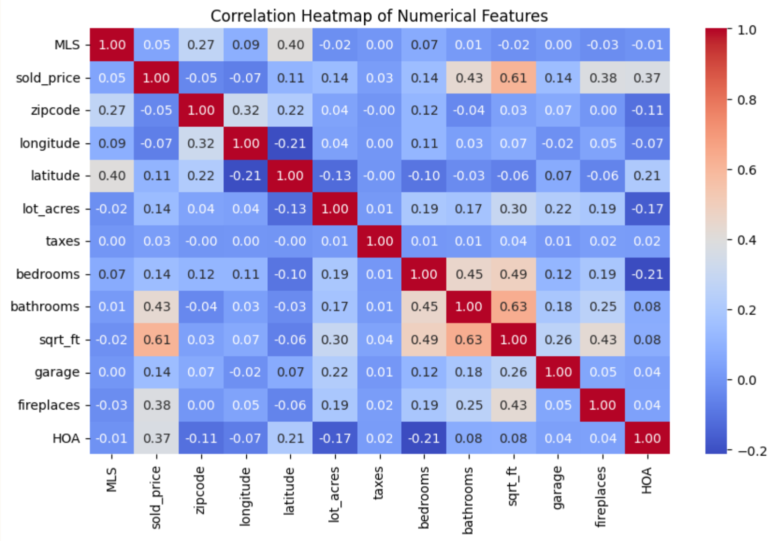


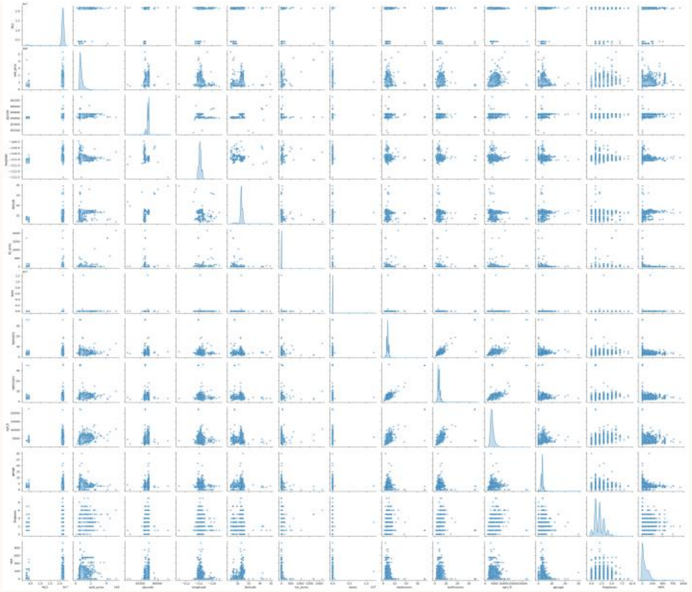


VISUALIZATIONS





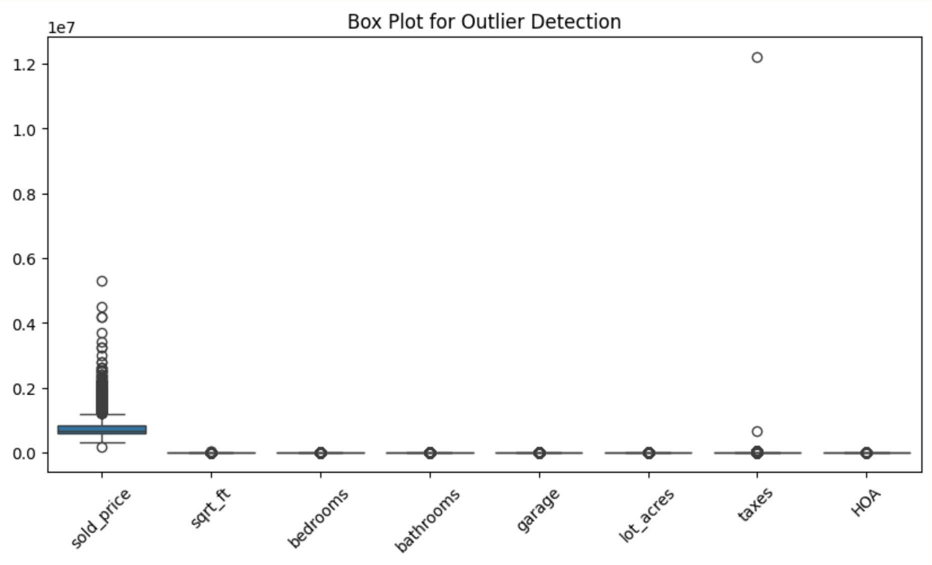


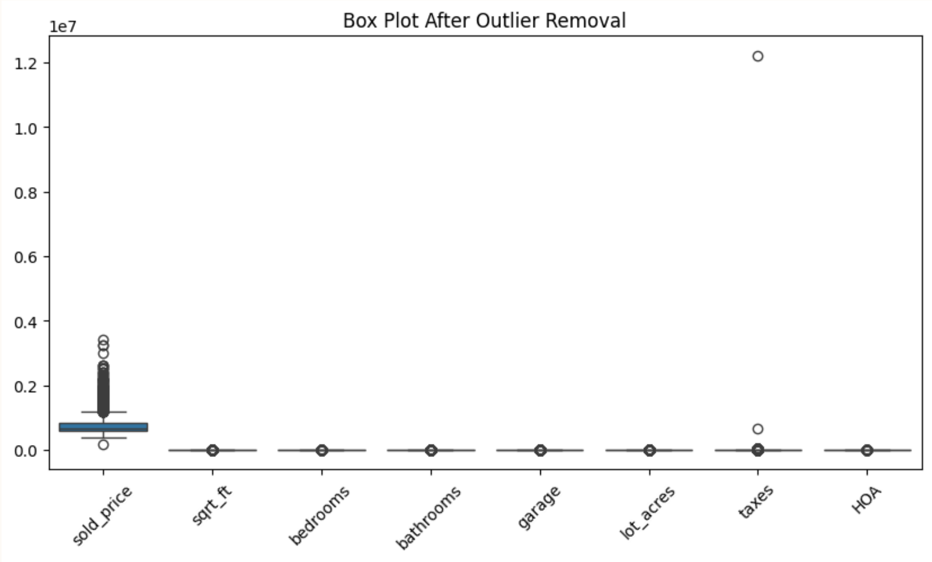


```
[ ]  # Generate Pair Plots
     sns.pairplot(data, diag_kind='kde', plot_kws={'alpha': 0.5})
     plt.show()
```

REMOVING OUTLIERS

- **Identified Outliers** using **IQR method** for all numerical features.
- **Feature with Most Outliers:** `lot_acres` (575 outliers).
- **Outliers Removed** using IQR range:
- **Lower Bound:** `Q1 - 1.5 * IQR`
- **Upper Bound:** `Q3 + 1.5 * IQR`
- Filtered out values outside this range.
- **Replotted Box Plot** after removal to verify the change.
- **Effect:** Reduced extreme values, improving model performance.





4.3 FEATURE ENGINEERING

Creating new features based on existing ones.

1. **price_per_sqft** feature : helps to normalize property prices based on size.

```
data['price_per_sqft'] = data['sold_price'] / data['sqrt_ft']
```

2. **tax_price_ratio** feature : high property taxes can affect buyer's decisions of buying the house

```
data['tax_price_ratio'] = data['taxes'] / data['sold_price']
```

3.Converts HOA into categories : **HOA_Category**

```
#3. Converts HOA into categories
def categorize_hoa(fee):
    if fee == 0:
        return "No HOA"
    elif fee <= 100:
        return "Low"
    elif fee <= 300:
        return "Medium"
    else:
        return "High"

data['HOA_category'] = data['HOA'].apply(categorize_hoa)
```

CONCLUSION AND FUTURE INSIGHTS

- After cleaning the data, handling missing values, conducting exploratory data analysis (EDA), removing outliers, creating visualizations, and performing feature engineering, I successfully obtained a clean housing dataset.
- data.to_csv("cleaned_house_data.csv", index=False)
- print("Cleaned dataset saved as cleaned_house_data.csv")
- This clean housing data now contains of **5000 rows and 19 columns** due to new addition of features as part of feature engineering done prior.
- This clean dataset can be used to train ML models like Naïve Bayes and KNN for both classification and regression tasks, with the target variable being **sold_price**. Subsequently, we can evaluate the accuracy and performance of the models.
- Further we can Implement real-time data updates by incorporating recent housing sales and features to continuously retrain the model and improve prediction accuracy.

THANK YOU